1 Linear algebra and set review

Proof formats **(Set inclusion)** Prove $P = Q \implies$ Prove $(P \subseteq Q)$ and $(Q \subseteq P)$. **(Contradiction)** Prove $A \Rightarrow B \implies$ Assume A and $\neg B$ and find contradiction. (**Contrapositive**) Prove $A \Rightarrow B \implies$ Prove $\neg B \Rightarrow \neg A$. **(Induction)** Prove $A(n) \implies$ *Base:* prove $A(0)$, *Assumption:* assume $A(n)$; *Induction:* derive $A(n+1)$.

Matrix formats $A = \begin{bmatrix} A_1 & A_2 & \dots & A_n \end{bmatrix} = \begin{bmatrix} a_1' & \dots & a_m' \end{bmatrix}^T, \quad Ax = \sum_i A_i x_i = \begin{bmatrix} a_1'x & \dots & a_m'x \end{bmatrix}^T$

$A \in \mathbb{R}^{n \times n}$ **invertible** $\iff \operatorname{rank}(A) = n \iff$ the columns of A are linearly indepenent $\iff \forall\, b,\ Ax = b$ has unique solution.

Polyhedron A set that can be described in the form $P = \{x \in \mathbb{R}^n \mid Ax \geq b\}$. To **test boundedness** use contradiction: Suppose $\exists d \neq 0$ such that $x + \lambda d \in P\ \forall \lambda \geq 0$, usually arriving at contradiction that $d = 0$.

Distance The shortest path from point x_0 to plane $a^T x = b$ is orthogonal to the plane. Write as $x = x_0 - t * a \implies b = a^T x = a^T(x_0 - t * a) = a^T x_0 - t \left\|a\right\|_2^2 \implies t = (b - a^T x_0)/\left\|a\right\|_2^2$. The signed displacement is $(x_0 - x) = t * \left\|a\right\|_2 = (b - a^T x_0)/\left\|a\right\|_2$

Projection $\Pi_k(S) = \{\pi_k(x) \mid x \in S\}$ where $\pi_k(x) = \pi_k(x_1, \dots, x_n) = (x_1, \dots, x_k)$. S nonempty $\iff$ $\Pi_k(S)$ nonempty. **Corrolary** For $P \subset \mathbb{R}^{n+k}$ a polyhedron, then $\{x \in \mathbb{R}^n \mid \exists y \in \mathbb{R}^k$ with $(x, y) \in P\}$ is a polyhedron.

Fourier-Motzkin elimination Tradeoff dimension decrease for constraints increase. **Elimination algorithm:** Lift optimization problem: $\max_{\{x \mid Ax \geq b\}} c^T x \Rightarrow P' = \{(x, x_0) \mid Ax \geq b, c^T x = x_0\}$. Use elimination to arrive at $Q = \{x_0 \mid \alpha_i x_0 \leq \beta_i\}$; the optimization becomes $\min_{\{x_0 \mid x \in Q\}}$.

$$P = \{x \in \mathbb{R}^n \mid a_i'x \geq b_i, i \in [m]\} = \left\{x \in \mathbb{R}^n \mid a_{in}x_n \geq -\sum_j a_{ij}x_j + b_i, i \in [m]\right\}, \qquad Q = \left\{x \in \mathbb{R}^{n-1} \mid \begin{cases} d_j + f_j'\overline{x} \geq d_i + f_i'\overline{x} & a_{jn} < 0, a_{jn} > 0 \\ 0 \geq d_k + f_k'\overline{x} & a_{kn} = 0 \end{cases}\right\} = \Pi_{n-1}(P)$$

Convexity A **Convex function** $f(\lambda x + (1-\lambda)y) \leq \lambda f(x) + (1-\lambda)f(y),\ \forall f : \mathbb{R}^T \mapsto \mathbb{R},\ \forall x, y \in \mathbb{R}^n,\ \forall \lambda \in [0,1]$. A **Convex set** $S \subset \mathbb{R}^n$ such that $\lambda x + (1-\lambda)y \in S, \forall x, y \in S, \lambda \in [0,1]$. A **convex combination** of a finite number of elements of convex set, S also belongs to that set. **(Proof with induction)** Base case: given for 2 elements by definition. Assume step k, then

$$\sum_{i=1}^{k+1} \lambda_i x^i = \lambda_{k+1} x^{k+1} + (1 - \lambda_{k+1}) \sum_{i=1}^{k} \frac{\lambda_i}{1 - \lambda_{k+1}} x^i \qquad \text{with } \lambda_i \geq 0, \sum_i \lambda_i = 1,\ \left(\sum_{i=1}^{k} \frac{\lambda_i}{1 - \lambda_{k+1}} x^i\right) \in S$$

Affine subspace $y \in S = \{x \mid Ax = b\} \iff (x - x^0) \in S_0 = \{y \mid Ay = 0\},\ \forall x \in S$ and for fixed $x^0 \in S$. **Subspace dimension** Let $S = \{x \mid Ax \geq b\}$ and fix any x^0 with $Ax^0 \geq b$. Then $S = \{x^0 + y \mid y \in S_0\}, \quad S_0 = \{y \mid Ay \geq 0\}$. The affine hull of S is $\operatorname{aff}(S) = \{x_0 + \operatorname{span}(x - x_0 : x \in S)\}$. Its dimension is $\dim(S) = \dim(\operatorname{aff}(S)) = n - \operatorname{rank}(A_{I_=})$, where $I_= = \{i \mid a_i^T x = b, \forall x \in S\}$.

2 Linear programming

Extreme points $\iff$ vertices $\iff$ basic feasible solutions: **Extreme point** $x \in P$ such that $\nexists y, z \in P,\ \lambda \in [0,1]$ such that $x = \lambda y + (1-\lambda)z$ and $x \neq y, z$. **Vertex** $x \in P$ such that $\exists c$ where $c'x < c'y,\ \forall\, y \in P, y \neq x$ **Basic feasible solution** A **basic solution** satisfies $x \in \mathbb{R}^n$ such that i) all equality constraints of P are active, ii) of all active constraints of P, n are linearly independent. A **basic feasible solution** is a basic solution with $x \in P$

Extreme point $\Rightarrow$ BFS: Let $I = \{i \mid a_i'x = b_i\}$. Suppose $x \in P$ not a BFS, then normals $\{a_i\}_{i \in I}$ do not span $\mathbb{R}^n \implies \exists\, d \neq 0 \in \mathbb{R}^n$ such that $a_i'd = 0 \forall\, i \in I$. Consider points $y = x + \epsilon d,\ z = x - \epsilon d$. For $i \in I, a_i'y = a_i'x + \epsilon a_i'd = b_i$ so constraints remain active. For $i \notin I$ we have $a_i'x > b_i$. Let $\bar{\epsilon} = \min\left\{\frac{a_i'x - b_i}{|a_i'd|} \mid i \notin I,\ a_i'd \neq 0\right\} > 0$. Then $\forall \epsilon \in [0, \bar{\epsilon}], i \notin I, a_i'y = a_i'x + \epsilon a_i'd \geq a_i'x - |\epsilon a_i'd| > a_i'x - (a_i'x - b_i) = b_i$. Hence $y, z \in P$. Since $d \neq 0, y \neq z,\ x = \frac{1}{2}(y + z)$, a nontrivial convex combination of two distinct feasible points. Contradiction $\square$

BFS implies $x_{j \notin I} = 0$: Consider $Ax = b, x \geq 0$ with $A \in \mathbb{R}^{m \times n}$ with m linearly independent rows. x a basic solution $\iff Ax = b$ and $\exists$ m indices B indices such that A_B are linearly independent. Then $Ax = A_B x_B = b$ and if $j \notin B \rightarrow x_j = 0$.

Existence of extreme points For nonempty $P = \{x \mid Ax \geq b\}$ the following are equivalent: (a) P has at least one extreme point, (b) P does not contain a line, (c) $\exists n$ linearly independent rows in A. **(b)$\Rightarrow$(a)** Start at any $x \in P$. If x a BFS, we're done. Else, let $I = \{i \mid a_i'x = b\}$ and note $\exists d \neq 0$ such that $a_i'd = 0 \forall i \in I$. No line $\Rightarrow \exists \lambda, j \notin I$ such that $a_j'(x + \lambda d) = b_j$. Note $j \notin I \Rightarrow a_j'x \neq b_j \Rightarrow a_j'd \neq 0$ and $i \in I \Rightarrow a_i'd = 0$ by construction. Together, a_j cannot be a linear combination of $\{a_i\}_{i \in I}$. **Corrolaries:** Bounded polyhedron has at least one BFS. $\{x \mid x \geq 0\}$ does not contain a line. Standard form polyhedron does not contain a line since $P \subseteq \{x \mid x \geq 0\}$

Existence of optimal extreme point For $\min_{\{x \in P\}} c'x$ and P has at least one extreme point. Then optimal cost is either $-\infty$ or $\exists$ extreme point that's optimal. **(Proof)** for some $x \in P$ with rank $k < n$, we show we can construct $y \in P$ with rank $k + i$ and $c'y \leq c'x$. Begin with $I = \{i \mid a_i'x = b_i\}$, which by rank, $k \Rightarrow \exists d \neq 0$ such that $a_i'd = 0 \forall i \in I$. WLOG, $c'd \leq 0$ since we can always negate d. Consider $y = x + \lambda d \Rightarrow a_i'y = b_i\, \forall\, i \in I$. If $y \in P\ \forall \lambda > 0 \Rightarrow$ optimal cost would be $-\infty$. Else, this half-line exits P. At the point of exit, we have $\lambda^*, j \notin I$ such that $a_j'(x + \lambda^* d) = b_j$. Note $c'd \leq 0 \Rightarrow c'y \leq c'x$. Here we have produced new y whose cost has not increased, but whose rank has. The process can be repeated until rank is n

Existence of unique optimum $\min_{\{x \mid Ax = b, x \geq 0\}} c^T x$, x^* optimal, B, N the sets of basic and nonbasic indices, $I \subseteq N = \{i \in N \mid \bar{c}_i = 0\}$. x^* unique optimal $\iff$ I empty. **(Proof sketch $\Leftarrow$)** Contradiction. Suppose $\exists y \neq x^*,\ c^T y = c^T x^*$. Let $d = y - x^* \Rightarrow Ad = 0 \Rightarrow d_B = -A_B^{-1} A_N d_N$. Notice $0 = c^T d = c_B^T d_B - c_N^T d_N = \bar{c}_N^T d_N$. $I = \emptyset$ and x^* optimal $\implies \bar{c}_N = 0 \implies d_N = 0 \implies d_B = 0 \implies y = x$ a contradiction $\square$

Degeneracy $x \in \mathbb{R}^n$ of P degenerate $\iff$ some basic variable is zero $\iff$ $\exists$ another basis that produces the same BFS $\iff$ $> n$ constraints are active at x. In standard form, $supp(x) < m$. Algorithm consequence: Possible stalling ($\bar{c}_j < 0$ and $x_j = 0$ allows for removing variable without affecting cost) Possible cycling.

Convert any problem to standard form WLOG, all rows of A are linearly independent (prove using two-way containment)

$$\max_{\{x|a_i'x \geq b_i, a_j'x \leq b_j\}} c'x \implies \max_{\{x|a_i'(x^+-x^-) \geq b_i, a_j'(x^+-x^-) \leq b_j, x^+ \geq 0, x^- \geq 0\}} c'x \implies \min_{\{x|a_i'x - s_i = b_i\ a_j'x + s_j = b_j, x, s \geq 0\}} -c'x$$

Problems involving absolute value At optimal, for each i we must have either $x_i^+ = 0$ or $x_i^- = 0$. Otherwise, we could change both by same amount, preserve feasibility, and reduce cost.

$$\min_{\{x|Ax \geq b\}} \sum_i c_i|x_i| \implies \min_{\{x|Ax \geq b, x_i \leq z_i, -x_i \leq z_i\}} \sum_i c_i' z_i \quad \text{or} \quad \min_{\{x|A(x^+-x^-) \geq b, x_i^+, x_i^- \geq 0\}} \sum_i c_i'(x_i^+ + x_i^-)$$

3 Simplex method

Feasible direction d such that, for $x \in P$, $\exists \theta > 0$ where $x + \theta d \in P$. At a BFS with basic/nonbasic indices, (B, N), we construct a search direction that guarantees feasibility: (i) $d_j = 1, j \in N$, (ii) $d_k = 0, k \in N \setminus \{j\}$, (iii), $d_B = -B^{-1}A_j$ since $A(x + \theta d) = b \Rightarrow 0 = Ad = \sum_{i=1}^n A_i d_i = \sum_{b \in B} A_b d_b + A_j = Bd_B + A_j$

Reduced cost vector For entering variable $j, \bar{c}_j = c'd = c_j + c_B^T d_B = c_j - c_B^T B^{-1} A_j$. For all directions, $\bar{c} = c - c_B^T B^{-1} A \in \mathbb{R}^n$
If $\bar{c} \geq 0 \Rightarrow x$ optimal: Define $d = y - x$,

$$y, x \in P \implies 0 = Ad = Bd_B + Nd_N \implies \mathbf{d_B = -B^{-1}Nd_N}$$
$$\mathbf{c^T d} = c_B^T d_B + c_N^T d_N = -c_B^T(B^{-1}Nd_N) + c_N^T d_N = (c_N^T - c_B^T B^{-1} N) d_N = \mathbf{\bar{c}_N^T d_N}$$

x BFS and y feasible $\implies x_N = 0, y_N \geq 0 \implies d_N = y_N - 0 \geq 0 \implies c^T d = \bar{c}_N^T d_N \geq 0 \implies c'(y - x) \geq 0 \implies c'y \geq c'x$, so x optimal.
(Corrolary) Basis B optimal if (a) $B^{-1}b \geq 0$ and (b) $\bar{c} \geq 0$.

Feasible and optimal θ Want $\theta^* = \max\{\theta \geq 0 \mid x + \theta d \in P\}$. Note if $d \geq 0 \implies x + \theta d \geq 0\ \forall \theta \geq 0$ and if $c^T d < 0 \implies \theta^* = \infty$. Otherwise, for $d_i < 0$, $x_i + \theta d_i \geq 0 \Rightarrow \theta \leq -x_i/d_i$. $\theta^* = \min_{\{i|d_i<0\}}(-x_i/d_i)$. Note for $d < 0$, $-x_i/d_i > 0$

Entering variable and non-leaving basic variables are linearly independent Call new basis $\bar{B}$ and suppose they were linearly dependent $\Rightarrow \exists \lambda \neq 0$, such that $\lambda' A_{\bar{B}} = 0 \Rightarrow \lambda' A_B^{-1} A_{\bar{B}} = 0$. But since $A_B^{-1} A_B = I \Rightarrow A_B^{-1} A_{\bar{B}} = [e_1, \dots A_B^{-1} A_j, \dots e_m]$. The ℓth entry is $-d_{B_\ell} \neq 0$ and all other columns are 0 in the ℓth entry, thus $A_B^{-1} A_j$ is linearly independent from $A_B^{-1} A_{B \setminus \{\ell\}}$.

Simplex iteration (1) Start with BFS x and associated basis indices, B. (2) Compute the reduced cost vector $\bar{c}$. If $\bar{c} \geq 0 \Rightarrow x$ optimal and terminate. Else choose $j : \bar{c}_j < 0$. (3) Compute $d_B = -A_B^{-1} A_j$. If $d \geq 0 \Rightarrow \theta^* = \infty$ since $\bar{c}_j = c'd < 0$;terminate with optimal cost $-\infty$. Else choose $\theta^* = \min_{\{i \in B | d_i < 0\}}(-x_i/d_i)$ (4) Identify leaving index $\ell : \theta^* = (-x_\ell/d_\ell)$ and form a new basis replacing A_{B_ℓ} with A_j. New BFS, y has $y_j = \theta^*$, $y_B = x_B + \theta^* d$, $y_i = 0$ otherwise. (4) Add to each non-zeroth row a multiple of the ℓth row so $-d_\ell = [A_B^{-1} A_j]_\ell = 1$ and $-d_{i \neq \ell} = [A_B^{-1} A_j]_{i \neq \ell} = 0$

Simplex terminates for nondegenerate problems At each step, we move θ^* along $d : c'd < 0$. Never revisit BFS.

Initializing simplex method Consider standard form problem $\min_{\{x|Ax=b, x \geq 0\}} c'x$. WLOG, $b \geq 0$. Construct new problem $\min_{\{x,y|Ax+y=b, x \geq 0, y \geq 0\}} \mathbf{1}'y$. This problem has a known BFS, namely $(x, y) = (0, b)$. Solving this problem returns BFS x for the original problem. **Infeasibility detection:** If $\mathbf{1}^T y^* > 0 \implies \nexists x \geq 0 : Ax = b$.

Anticycling **Lexicographic pivoting:** Always keep non-zero rows sorted lexicographically. For each row i with $d_i < 0$, normalize row i by dividing by d_i. Always pivot on lexicographically smallest

Revised simplex Form matrix $[A_B^{-1} \mid A_B^{-1} A_j]$, perform row operations, Q, to obtain $[\bar{A}_B^{-1} \mid e_\ell]$; solves A_B^{-1}, in $O(m^2)$.

Tableau implementation Here the efficiency comes from maintaining and updating $A_B^{-1}[b \mid A]$. The tableau structure is

$$\left[\begin{array}{c|c} -c_B^T A_B^{-1} b & c^T - c_B^T A_B^{-1} A \\ \hline A_B^{-1} b & A_B^{-1} A \end{array}\right] = \left[\begin{array}{c|ccc} -c^T x & \bar{c}_1 & \dots & \bar{c}_n \\ \hline x_B & -d^{(1)} & \dots & -d^{(n)} \end{array}\right]$$

Step		Iteration 1	Iteration 2
1.	Choose pivot column, l, based on $\bar{c}^T$ (lexicographically, or smallest $\bar{c}_j$)	$\begin{bmatrix} 0 & -2 & -1 & 0 & 0 \\ 2 & 1 & -1 & 1 & 0 \\ 6 & 1 & 1 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 6 & -1 & 0 & 0 & 1 \\ 8 & 2 & 0 & 1 & 1 \\ 6 & 1 & 1 & 0 & 1 \end{bmatrix}$
2.	Choose pivot row, i, using ratio test ($\{i \in B \mid d_i^{(\ell)} < 0, \text{argmin}(-x_i/d_i^{(\ell)})\}$)	$\begin{bmatrix} 0 & -2 & -1 & 0 & 0 \\ 2 & 1 & -1 & 1 & 0 \\ 6 & 1 & 1 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 6 & -1 & 0 & 0 & 1 \\ 8 & 2 & 0 & 1 & 1 \\ 6 & 1 & 1 & 0 & 1 \end{bmatrix}$
3.	Apply row operations to get $d_i^{(\ell)} = 1$ and $d_{k \neq i}^{(\ell)} = 0$	$\begin{bmatrix} 0 & -2 & -1 & 0 & 0 \\ 8 & 2 & 0 & 1 & 1 \\ 6 & 1 & 1 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 6 & -1 & 0 & 0 & 1 \\ 4 & 1 & 0 & 1/2 & 1/2 \\ 2 & 0 & 1 & -1/2 & 1/2 \end{bmatrix}$
4.	Apply row operations to set $\bar{c}_l = 0$	$\begin{bmatrix} 6 & -1 & 0 & 0 & 1 \\ 8 & 2 & 0 & 1 & 1 \\ 6 & 1 & 1 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 10 & 0 & 0 & 1/2 & 3/2 \\ 4 & 1 & 0 & 1/2 & 1/2 \\ 2 & 0 & 1 & -1/2 & 1/2 \end{bmatrix}$

4 Duality

Lagrangian Consider standard form problem $\min_{\{x|Ax=b, x \geq 0\}} c'x$. The lagrangian is $L(x, p) = c'x + p'(b - Ax)$ and the Dual function is $g(p) = \min_{x \geq 0} c'x + p'(b - Ax) = p'b + \min_{x \geq 0}(c' - p'A)x$. **(Saddlepoint)** A pair $(x^*, p^*), x^* \geq 0$ are primal-dual optimal (standard-form) $\iff$ they are a saddlepoint: $L(x^*, p) \leq L(x^*, p^*) \leq L(x, p^*)\ \ \forall x \geq 0, p$. **(Dual problem)** The solution to the inner minimization is 0 if $\mathbf{c' - p'A \geq 0}$ and $-\infty$ otherwise, so to maximize $g(p)$ the Dual Problem is $\max_{\{p|p'A \leq c'\}} p'b$.

Sign convention In a **minimization problem** with **natural inequalities** $Ax \geq b$, the pairs $(p, b - Ax)$ and $(x, c - p^T A)$ have opposite inequality directions. Constraint is violated when $Ax < b$. In minimization, we want the penalty to increase the Dual function, which is achieved with $p \geq 0$ through $p'(b - Ax)$. In a **maximization problem** with **natural inequalities** $Ax \leq b$, the pairs $(p, Ax - b)$ and $(x, c - p^T A)$ have opposite inequality directions. Intuition: $x_j(c_j - p^T A)$ encodes reduced costs. **Equality constraints** correspond to free dual variables and always consider $b - Ax$.

Common dual-primal pairs $\left((P) : \min_{\{x|Ax=b, x\geq 0\}} c^T x \iff (D) : \max_{\{p|p \text{ free}, p^T A \leq c\}} p^T b\right)$
$\left((P) : \min_{\{x|Ax\geq b\}} c^T x \iff (D) : \max_{\{p|p\geq 0, p^T A = c\}} p^T b\right)$
$\left((P) : \min_{\{x|Ax\geq b, x\geq 0\}} c^T x \iff (D) : \max_{\{p|p\geq 0, p^T A \geq c\}} p^T b\right)$
$\left((P) : \max_{\{x|Ax\leq b, x\geq 0\}} c^T x \iff (D) : \min_{\{p|p\geq 0, A^T p \geq c\}} b^T p\right)$
$\left((P) : \min_{\{x|A_1 x\geq b_1, A_2 x = b_2, x\geq 0\}} c^T x \iff (D) : \max_{\{p_1, p_2 | p_1 \geq 0,\ p_2 \text{ free}, p_1^T A_1 + p_2^T A_2 \leq c\}} p_1^T b_1 + p_2^T b_2\right)$

Dual simplex In Tableau, always keep Dual constraints, $\bar{c} = c^T - p^T A \geq 0$, and pivot to make primal constraint $A_B^{-1} b \geq 0$.

$$\left[\begin{array}{c|c} -p^T b & c^T - p^T A \\ \hline A_B^{-1} b & A_B^{-1} A \end{array}\right] = \left[\begin{array}{c|c} -c_B^T A_B^{-1} b & c^T - c_B^T A_B^{-1} A \\ \hline A_B^{-1} b & A_B^{-1} A \end{array}\right] = \left[\begin{array}{c|c} -c^T x & \bar{c} n \\ \hline A_B^{-1} b & A_B^{-1} A \end{array}\right]$$

Dual simplex iteration (1) Start with basic solution x and associated basis indices, B, and $\bar{c} \geq 0$. (2) Compute $A_B^{-1} b$. If $A_B^{-1} b \geq 0 \Rightarrow x$ optimal and terminate. Else choose $\ell : A_B^{-1} b_\ell < 0$. (3) Compute $v^{(\ell)} = [A_B^{-1} A]_\ell$. If $v^{(\ell)} \geq 0 \Rightarrow$ problem is unbounded; terminate with optimal cost ∞. Else choose $\gamma^* = \min_{\{i | v_i^{(\ell)} < 0\}} (\bar{c}_i / |v_i|)$. (4) Identify leaving index $j : \gamma^* = (\bar{c}_j / |v_j|)$ and form a new basis replacing A_{B_ℓ} with A_j. New BFS, y has $y_j = \theta^*,\ y_B = x_B + \theta^* d,\ y_i = 0$ otherwise. (5) Add to each non-zeroth row a multiple of the ℓth row so $v_j^{(\ell)} = 1$ and $v_j^{(i \neq \ell)} = 0$

Dual degeneracy For basis, B, a basic solution in the dual is $p^T = c_B^T A_B^{-1}$. p^T degenerate $\iff \exists$ some nonbasic index j with reduced cost $\bar{c}_j = 0 \implies$ i.e., multiple primal BFS correspond to the same dual solution. **Dual-primal link** Dual degenerate $\implies$ multiple primal optimal BFS yield same x^*. Primal degenerate $\implies$ multiple dual feasible basis yield same p^*. **Degenerate dual $\not\Rightarrow$ degenerate primal**: $(P) : \min_{\{x|x_1, x_2 \geq 0, x_1 + x_2 \geq 0\}} x_2,\ (D) : \min_{\{p|p\geq 0, p\geq 0, p\geq 1\}} p0$

Weak duality Let (x, p) feasible $(P), (D)$, then $p'b \leq c'x$. **(Proof)** Let $u_i = p_i(a_i' x - b_i),\ v_j = (c_j - p' A_j) x_j$. By construction, $\text{sign}(a_i' x - b) = \text{sign}(p_i)$ and $\text{sign}(c_j - p' A_j) = \text{sign}(x_j)$ So $u_i, v_j \geq 0\ \forall i, j$. Now

$$0 \leq \sum_i u_i + \sum_j v_j = p' A x - p' b + c' x - p' A x = c' x - p' b \implies p' b \leq c' x$$

Strong duality If $(P), (D)$ have optimal solutions, the respective costs are equal: $c'x = p'b$.

Strong duality proof using simplex Consider primal-dual pair, $(P) : \min_{\{x|Ax=b, x\geq 0\}} c^T x,\ (D) : \max_{\{p|p \text{ free}, p^T A \leq c\}} p^T b$. We can find x^* optimal with basis A_B using the simplex algorithm. At the optimum, $\bar{c}^T = c^T - c_B A_B^{-1} A \geq 0$. Let $p^* = c_B A_B^{-1}$ and note $p^{*T} A \leq c^T \implies p^*$ feasible. Also $p^{*T} b = c_B^T A_B^{-1} b = c_B^T x_B^* = c^T x^*$ so p^* optimal with equal optimal cost $\square$

Strong duality proof using Farkas' lemma Consider primal-dual pair, $(P) : \min_{\{x|Ax\geq b\}} c'x,\ (D) : \min_{\{p|p\geq 0, p' A = 0\}} p'b$ and assume primal optimal x. Let $I = \{i \mid a_i' x = b_i\}$. Choose d such that $a_i d \geq 0\ \forall i \in I \Rightarrow a_i'(x + \epsilon d) \geq a_i x = b_i\ \forall i \in I$. For $j \notin I$ and ϵ sufficiently small, $a_j'(x + \epsilon d) > b_i \Rightarrow x + \epsilon d$ feasible solution. By optimality of $x \Rightarrow c'd \geq 0$. By Farkas' lemma, $\exists p_i \geq 0, i \in I$ such that $c = \sum_{i \in I} p_i a_i$. For $j \notin I$, let $p_j = 0$. Now we have $p \geq 0,\ p' A = c'$. Also, $p'b = \sum_{i \in I} p_i b_i = \sum_{i \in I} p_i a_i x_i = c'x$ $\square$

Primal-dual outcome possibilities $(P)/(D)$ unbounded $\Rightarrow (D)/(P)$ infeasible. **(Proof)** (P) unbounded $\Rightarrow c'x = -\infty \Rightarrow p'b \leq -\infty$ by weak duality; impossible. (P) finite optimum $\iff$ (D) finite optimum by strong duality. $(P)/(D)$ infeasible $\Rightarrow (D)/(P)$ infeasible *or* unbounded.

Complementary slackness x, p optimal solutions for $(P), (D) \iff p_i(a_i' x - b_i) = 0\ \forall i, (c_j - p' A_j) x_j = 0\ \forall j$. Recall in the proof of **Weak duality** we defined u_i, v_j and showed $u_i \geq 0, v_j \geq 0, c'x - p'b = \sum_i u_i + \sum_j v_j$. **(Proof $\Rightarrow$)** By **Strong duality**, $c'x = p'b \implies \sum_i u_i + \sum_j v_j = 0 \implies u_i = v_j = 0 \forall i, j$. **(Proof $\Leftarrow$)** if $u_i = v_j = 0\ \forall i, j \implies c'x - p'b = 0 \implies c'x - p'b$.

Strict complementary slackness Given primal-dual pair $(P) : \min_{\{x|Ax=b, x\geq 0\}} c^T x, (D) : \max_{\{p|p^T A \leq c\}} p^T b$, both with optimal solutions, then $\exists$ optimal $x \geq 0, p$ such that $\forall j$ either $x_j > 0$ or $p^T A_j < c_j$. **(Proof sketch)** (i) first prove $\exists p : p^T A \leq c, p^T A_j < c_j$ in case where $x_j = 0 \forall x^{*(i)}$. Set up primal-dual problems $(P') : \min_{\{x|Ax=b, x\geq 0\}} c^T x - \delta e_j^T x, (D) : \max_{\{p|p^T A \leq c - \delta e_j\}} p^T b$. Consider reduced cost vector $\bar{c}' = (c - \delta e_j) - c_B B^{-1} A \implies \delta \leq c_j - c_B B^{-1} A_j$. Reason that for sufficiently small δ, same x^* is optimal; note new constraint of (D'). (ii) Then consider the sets of optimal solutions and corresponding sets of indices:

$$\mathcal{X} = \{x \mid Ax = b, x \geq 0, c^T x = val(P)\} \qquad \mathcal{P} = \{p \mid p^T A \leq c, p^T b = val(D)\}$$
$$S = \{j \mid \exists x \in \mathcal{X}, x_j > 0\} \qquad S^C = \{j \mid \exists x_j = 0\ \forall x \in \mathcal{X}\}$$

Define $\bar{x} = \frac{1}{|S|} \sum_{k \in S} x^{(k)}$, $\bar{p} = \frac{1}{|S^C|} \sum_{k \in S^C} p^{(k)}$. Prove feasiblity and optimality. Since S, S^C partition the indices, we have $\forall j \in [n]$ either $x_j > 0$ or $p^T A_j < c_j$

5 Farkas' lemma

Intuition Consider standard form constraints, $Ax = b, x \geq 0$. Suppose $\exists p$ such that $p' A \geq 0',\ p'b < 0$. Then $\forall x \geq 0,\ p' A x \geq 0$. But since $p'b \leq 0$ we have $p' A x \neq p'b \Rightarrow Ax \neq b\ \forall x \geq 0$.

Farkas' lemma For A, b: $\exists x$ such that $x \geq 0, Ax = b \iff \nexists p$ such that $p'A \geq 0', p'b < 0$. (**Alternatives:** (i) For A, b: $\forall p$ such that $p'A \geq 0 \Rightarrow p'b \geq 0 \iff \exists x \geq 0$ such that $b = Ax$. (ii) For A, b, d: $\forall x$ such that $Ax \leq b \Rightarrow c'x \leq d \iff \exists p \geq 0$ such that $p'A = c'$ and $p'b \leq d$)

Farkas' lemma proof using Duality **(Proof (a)⇒ ¬(b))** $\exists x : x \geq 0, Ax = b \Rightarrow p'b = p'Ax \Rightarrow$ so $\nexists p$ such that $p'b < 0$ and $p'b = p'Ax \geq 0$. **(Proof ¬(a)⇒(b))** Consider primal-dual pair, $(P) : \max_{\{x|x\geq 0, Ax=b\}} 0'x$, $(D) : \min_{\{p|p'A\geq 0\}} p'b$. Not (a) $\rightarrow (P)$ infeasible $\Rightarrow (D)$ unbounded, since $p = 0$ is feasible. (D) unbounded $\Rightarrow \exists p$ feasible such that $p^T b < 0$.

Farkas' lemma proof using Separating hyperplanes **(Proof ¬(a)⇒(b))** Let $S = \{y \mid \exists x : y = Ax, x \geq 0\} = \{Ax \mid x \geq 0\}$, noting $b \notin S$. By Separating Hyperplane theorem, $\exists \mathbf{p} : \mathbf{p^T b} < \mathbf{p^T y}\ \forall y \in S$. Now $\forall \lambda > 0, \lambda A_i \in S \Longrightarrow p^T b < \lambda p^T A_i \iff p^T A > (1/\lambda) p^T b \Rightarrow \lim_{\lambda\to\infty} p^T A_i \geq 0 \Rightarrow \mathbf{p^T A \geq 0}$.

Farkas' lemma worked examples

- **(Statement)** Every vector satisfying $Ax \geq 0, x \geq 0$ also satisfies $x_1 = 0 \iff \exists p \geq 0$ such that $p^T A \leq 0, p^T A_1 < 0$.
  (Proof) Consider primal-dual pair, $(P) : \max_{\{x|x\geq 0, Ax\geq 0\}} -e_1^T x$, $(D) : \min_{\{p|p^T A \leq -e_1, p\geq 0\}} p^T 0$. **(Proof ⇒)** $val((P)) = 0 \Longrightarrow (D)$ feasible and there exists such a p. **(Proof ⇐)** $p^T A_1 < 0 \Rightarrow p^T A_1 \leq -1 \Rightarrow (D)$ feasible with $val((D) = 0) \Longrightarrow e_1^T x = 0 \Longrightarrow x_1 = 0$.
- **(Statement)** $\exists x \neq 0$ such that $Ax = 0, x \geq 0 \iff \nexists p$ such that $p^T A > 0$
  (Proof ⇒) Consider primal-dual pair, $(P) : \min_{\{x|x\geq 0, Ax=0\}} 0^T x$, $(D) : \max_{\{p|p \text{ free}, p^T A\leq 0\}} p^T 0$. (P) feasible $\Rightarrow (D)$ feasible and $p^T A \leq 0$. **(Proof ⇐)** Note $p^T A < 0 \Rightarrow p^T A \leq e$ because we can scale. Now consider primal-dual pair, $(P) : \min_{\{p|A^T p\geq e\}} 0^T p$, $(D) : \max_{\{x|x\geq 0, Ax=0\}} e^T x$. (P) feasible $\Rightarrow e^T x = 0$ and $x \geq 0 \Rightarrow x = 0$.
- **(Statement)** $Ax < 0, x$ feasible $\iff S = \{u \mid u^T A = 0, u \geq 0, \mathbf{1}^T u = 1\} = \emptyset$
  (Proof ⇐ using Farkas) $S = \{u \mid [A, \mathbf{1}]^T u = [\mathbf{0}^T, 1]^T, u \geq 0\}$. By Farkas $\Longrightarrow \exists\ p, q$ free such that $p^T A^T + q\mathbf{1}^T \geq \mathbf{0}^T$ and $p^T \mathbf{0} + q < 0 \Longrightarrow q < 0$ and $Ap > 0 \Longrightarrow Ap < 0$
  (Proof ⇐ 1 using Strong duality) Consider primal-dual pair $(P) : \min_{\{u|[A,\mathbf{1}]^T u=[\mathbf{0}^T,1]^T, u\geq 0\}} \mathbf{0}^T u$, $(D) : \max_{\{(p,q)|Ap+\mathbf{1}q\leq 0, (p,q) \text{ free}}$ (P) infeasible $\Longrightarrow (D)$ unbounded (since $(0, 0)$ feasible) $\Longrightarrow q > 0 \Longrightarrow Ap < 0$.
  (Proof ⇒ 2 using Strong duality) Note $Ax < 0 \iff Ax \leq -\mathbf{1}$. Now consider primal-dual pair $(P) : \max_{\{x|Ax\leq -\mathbf{1}\}} \mathbf{0}^T x$, $(D) : \min_{\{u|u\geq 0, u^T A=0\}} -\mathbf{1}^T u$. (P) feasible $\Longrightarrow (D)$ feasible $\Longrightarrow -\mathbf{1}^T u = 0$ by strong duality $\Longrightarrow S = \emptyset$ since requires $\mathbf{1}^T u = 1$

Separating hyperplane theorem $S \in \mathbb{R}^n$ nonempty, closed, convex and $z \notin S$. Then $\exists c \in \mathbb{R}^n$ such that $c'z < c'x\ \forall x \in S$. **(Proof sketch)** Fix $w \in S$, let $B = \{x \mid \|x - z\|_2 \leq \|w - z\|_2\}$, and let $D = S \cap B$. $\exists y \in D$ such that $\|y - z\|_2 \leq \|x - z\|_2\ \forall x \in D$. $\forall x \in S \setminus D \Rightarrow \|x - z\|_2 > \|w - z\|_2 \geq \|y - z\|_2$ so y minimizes $\|x - z\|_2 \forall x \in S$ (so y is closest element of S to z). Let $x \in S$. S convex $\Rightarrow y + \lambda(x - y) \in S\ \forall \lambda \in (0, 1]$.

$$\|y - z\|_2^2 \leq \|y + \lambda(x - y) - z\|_2^2 = \|y - z\|_2^2 + 2\lambda(y - z)^T(x - y) + \lambda^2 \|x - y\|_2^2 \overset{1/\lambda}{\Longrightarrow} 2(y - z)^T(x - y) + \lambda \|x - y\|_2^2 \geq 0 \overset{\lim_{\lambda\to 0}}{\Longrightarrow} (y - z)^T(x - y) \geq 0$$
$$\Longrightarrow (y - z)^T x \geq (y - z)^T y = (y - z)^T z + (y - z)^T(y - z) > (y - z)^T z \Longrightarrow c^T x > c^T z \qquad (c = y - z)$$

6 Cones and extreme rays

Definitions **(Cone)** $C \in \mathbb{R}^n$ if $\lambda x \in C\ \forall \lambda \geq 0, x \in C$. **(Note)** $0 \in C$. **(Polyhedral cone)** $\mathcal{C} = \{x \in \mathbb{R}^n \mid Ax \geq 0\}$. For $P = \{x \in \mathbb{R}^n \mid Ax \geq b\}, y \in P$, **(recession cone at y)** $RC_P(y) = \{d \in \mathbb{R}^n \mid y + \lambda d \in P,\ \forall \lambda \geq 0\}$. Further, $RC_P(y) = RC_P(z) \forall y, z \in P \Longrightarrow RC_P = \{d \in \mathbb{R}^n \mid Ad \geq 0\}$. For $Q = \{x \in \mathbb{R}^n \mid Ax = b, x \geq 0\}, RC_Q = \{d \in \mathbb{R}^n \mid Ad = 0, d \geq 0\}$. **(Extreme ray def.1)** $x \neq 0 \in C \subset \mathbb{R}^n$ is an extreme ray if there are $n - 1$ linearly independent constraints active at x. **(Extreme ray def.2)** An extreme ray of RC_p with nonempty P is an extreme ray of P.

Equivalences For polyhedral cone $\mathcal{C} \subset \mathbb{R}^n$ defined by $a_i' x \geq 0, i \in [m]$, the following are equivalent: 0 is an extreme point of $\mathcal{C} \iff \mathcal{C}$ does not contain a line $\iff \exists n$ vectors out of $a_{[m]}$ which are linearly independent

Finite optimal cost over cone The optimal cost of $\min_{\{x\in C=\{x|Ax\geq 0\}\}} c^T x = -\infty \iff \exists$ extreme ray $d \in C$ such that $c'd < 0$. **(Proof ⇐)** Trivial. If $\exists d \in C$ such that $c'd < 0$ then $\lambda d \in C \forall \lambda \geq 0 \Rightarrow \min_{d\in C} c'd = -\infty$. **(Proof ⇒)** From given we assume $\exists x \in C$ such that $c'x = -1$. Define $P = \{x \mid Ax \geq 0, c^T x = -1\} \subset C$. C pointed $\Rightarrow$ C does not contain a line $\Rightarrow P$ has at least 1 extreme point, call it d. d extreme $\Rightarrow n$ linearly active constraints, one of which must be $c^T x = -1$. The remaining $n - 1$ are from $x \in C$ so d is an extreme ray of C.

Finite optimal cost over polygon The optimal cost of $\min_{\{x\in P=\{x|Ax\geq b\}\}} c^T x = -\infty \iff \exists$ extreme ray $d \in RC_P$ such that $c'd < 0$. **(Proof ⇐)** Trivial. If $d \in RC_P, c'd < 0$, then $\forall x \in P, x + \lambda d \in P\ \forall \lambda \geq 0 \Rightarrow c'(x + \lambda d) = c'x + \lambda c'd$. As $\lambda \to \infty \Rightarrow \lambda c'd \to -\infty$. **(Proof ⇒)** Consider primal-dual pair $(P) : \min_{\{x\in P=\{x|Ax\geq b\}\}} c^T x$, $(D) : \max_{\{p\in Q=\{p|p\geq 0, p^T A=c\}\}} p^T b$. (P) unbounded $\Rightarrow (D)$ infeasible $\Rightarrow (D') : \max_{\{p\in Q\}} p^T \mathbf{0}$ infeasible since same set as (D). Dual of (D'), $(P') = \min_{\{x\in \tilde{P}=\{x|Ax\geq 0\}\}} c^T x$ is unbounded (since $x = 0$ feasible). From previous theorem, P contains no line $\Rightarrow \exists n$ linearly independent constraints $\Rightarrow \exists$ same n linearly independent constraints in $\tilde{P} \Longrightarrow \tilde{P}$ is a pointed cone. Since $\tilde{P} \subseteq P \Rightarrow val((P)) \leq val((P')) = -\infty$. By previous theorem, $\exists$ extreme ray $d \in \tilde{P}$ such that $c'd < 0$. Conclude by noting that $\tilde{P} = RC_P$ $\square$

Resolution theorem Let $P = \{x \mid Ax \geq b\}$ nonempty with at least one extreme point. Let $x^{[k]}$ be all extreme points, $w^{[r]}$ all extreme rays of P. Then $P = Q = \left\{\sum_i \lambda_i x^i + \sum_j \theta_j w^j \mid \lambda_i \geq 0, \theta_j \geq 0, \sum_i \lambda_i = 1\right\}$. **(Proof $P \subset Q$)** Contradiction. Consider problem $(P) : \max_{\{\lambda,\theta|z=X\lambda+W\theta,\ \mathbf{1}^T\lambda=1, \lambda,\theta\geq 0\}} \mathbf{0}$. By assumption, (P) infeasible $\rightarrow (D) : \min_{\{p|p^T X+q\mathbf{1}^T\geq 0, p^T W\geq 0\}} p^T z + q$ unbounded (since $(p, q) = (0, 0)$ is feasible) $\Rightarrow \exists (p, q)$ such that $p^T z + q < 0$, but constraints say $p^T X + q\mathbf{1}^T \geq 0^T$ and $p^T W \geq 0^T \Rightarrow p^T z \mathbf{1}^T < p^T X \Rightarrow \exists p : p^T z < p^T x^i\ \forall z \in P, \forall i \in [k]$, but each x^i an extreme point, so for each i this is a contradiction.